

Build Your First Interactive Data App

 **with Streamlit**

From Python to a Web App

AGENDA

What we'll cover today

01 What & why Streamlit

02 How it works + setup

03 Your first app + widgets

04 Live build: sales dashboard

05 Deploy to GitHub & the web

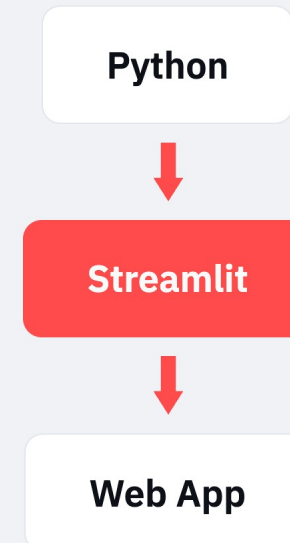
06 Recap, resources & Q&A

THE BASICS

What is Streamlit?

An open-source Python framework for building interactive web applications.

- Python-based — no HTML, CSS or JavaScript required
- Designed for data applications and dashboards
- Fast to build and prototype: script to app in minutes



THE BASICS

Why Streamlit?

SIMPLE

Build interfaces with plain Python

FAST

Create working prototypes quickly

INTERACTIVE

Users can interact with your data

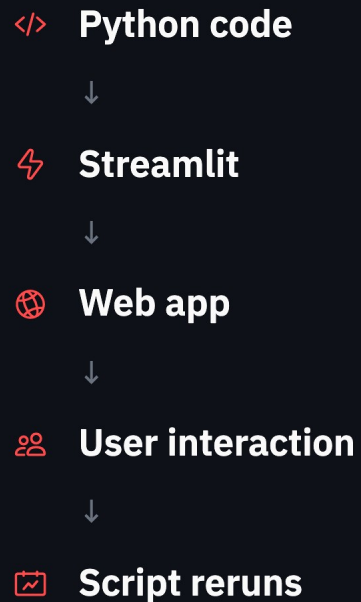
DATA-FRIENDLY

Works naturally with pandas & charts

Python → Data → Interface

THE BASICS

How does Streamlit work?



When a user interacts with the app, Streamlit reruns the whole Python script top to bottom and updates the interface. No manual frontend code required.

EXAMPLE

User moves a slider → Streamlit reruns → Python executes → UI updates

SETUP

Install Streamlit in two minutes

01

Create a virtual environment

02

Install streamlit & friends

03

Verify with the demo app



terminal

```
python -m venv .venv
source .venv/bin/activate
pip install streamlit pandas plotly
streamlit hello
```

SETUP

Your first Streamlit app

</> Write Python



⚡ Run Streamlit



🌐 Open in browser



app.py

```
import streamlit as st

st.title("My First Streamlit App")
st.write("Hello, Streamlit!")
```

```
$ streamlit run app.py
```

SETUP

Streamlit widgets

Text input

Your name: [...]

```
st.text_input()
```

Slider

Age: 25 ●—————

```
st.slider()
```

Selectbox

Language: Python ▼

```
st.selectbox()
```

User → Widget → **Streamlit reruns** → App updates

LIVE BUILD

Let's build a sales dashboard

From a CSV file to a live, interactive app — written together, step by step.

- 1 Load data
- 2 Filter data
- 3 Show KPIs
- 4 Create charts

STEP 1

Loading data



```
app.py

uploaded_file = st.file_uploader(
    "Upload your CSV file", type=["csv"]
)

df = pd.read_csv(uploaded_file)

st.dataframe(df)
```

STEP 2

Adding filters

Category filter

- ☒ Furniture
- ☒ Technology
- ☐ Office

All data



Select category



Filtered data

app.py

```
categories = df["Category"].unique()
selected = st.multiselect(
    "Select Category",
    categories,
    default=categories
)
filtered_df = df[df["Category"].isin(selected)]
```

STEP 3

KPIs & metrics

TOTAL SALES

\$125,430

↑ +12% vs. last month

AVERAGE SALES

\$2,450

↑ +8% vs. last month

PRODUCTS

52

↑ +5% vs. last month

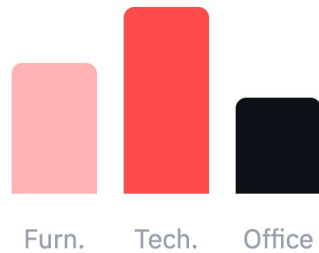
```
col1, col2, col3 = st.columns(3)

col1.metric("Total Sales", f"${total_sales:,.0f}", delta=f"{sales_change}%")
col2.metric("Average Sales", f"${avg_sales:,.0f}", delta=f"{avg_change}%")
col3.metric("Products", num_products, delta=f"{product_change}%")
```

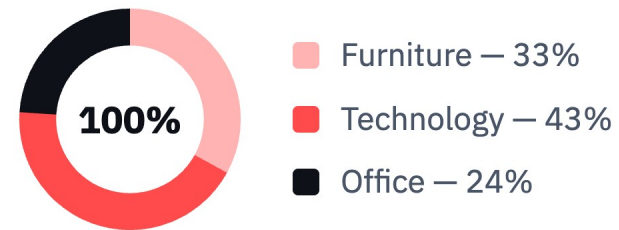
STEP 4

Create charts

Sales by category



Sales distribution



```
st.subheader("Sales by Category")
st.bar_chart(sales_by_category)

st.subheader("Sales Distribution")
fig = px.pie(df, names="Category", values="Sales")
st.plotly_chart(fig, use_container_width=True)
```

SHIP IT

From local app to the web



```
terminal

git init
git add .
git commit -m "first app"
git push origin main
```

Project structure

```
streamlit-pildora/
├─ app.py
├─ data/sales.csv
├─ requirements.txt
└─ README.md
```

Then connect the repo on share.streamlit.io — the app rebuilds on every push.

WRAP-UP

What we covered today

✓ What Streamlit is & why it's fast to build with

✓ Installing Streamlit & running your first app

✓ Core widgets: text input, slider, selectbox

✓ Loading & filtering a real CSV with pandas

✓ KPI cards with `st.metric` & charts with `plotly`

✓ Publishing to GitHub & Streamlit Community Cloud

WRAP-UP

Resources



Project repo on GitHub

`github.com/esrakesken/streamlit-pildora`



Official Streamlit documentation

`docs.streamlit.io`



Free hosting: Streamlit Community Cloud

`share.streamlit.io`

Connect your public GitHub repo and deploy in one click — no server, no card, always free.

Thank you!

Questions? Let's talk — and go build something with Streamlit.